ANN-MPC: An Adaptive Neural Network-Based Nonlinear Model Predictive Control Method for Robust Trajectory Tracking of Quadrotors

Chuixu Kong, Bin Wang, Liang He, Qinglong Jiang, and Chang Liu

***Abstract*—Quadrotor systems, despite their extensive deployment in diverse application domains, face substantial performance degradation due to three predominant factors: unmodeled aerodynamic phenomena, persistent wind disturbances, and unanticipated payload variations. These challenges collectively compromise precise trajectory tracking in dynamic environments. To address these limitations, this study introduces an integrated control architecture combining 1) an adaptive neural network disturbance observer (ANNDO) with guaranteed Lyapunov stability through carefully designed adaptive laws, and 2) a modified model predictive control (MPC) scheme enhanced with real-time disturbance compensation. The proposed framework demonstrates substantial performance improvements, achieving over 80% tracking accuracy enhancement in numerical simulations during agile flight, and over 50% improvement in physical experiments involving complex disturbance rejection, compared with baseline MPC.**

***Index Terms*—Adaptive neural network control, model predictive control, disturbance rejection.**

I. Introduction

A. Motivation

THE proliferation of quadrotor systems in mission-critical domains, including time-sensitive logistics operations [1], [2] and safety-contingent emergency response missions [3], has been propelled by recent advancements in embedded avionics and fusion perception. Ensuring high-precision trajectory tracking under disturbances is essential for quadrotors to maintain safe and high-performance flight in complex environments.

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Traditional control methods, such as PID and LQR, are widely used due to their simplicity and ease of implementation [4]. However, their performance markedly deteriorates when addressing highly nonlinear dynamics inherent in aerial systems. Research demonstrates that MPC effectively addresses the nonlinear coupling effects of quadrotors through receding horizon optimization with explicit constraint handling [5], [6]. Nevertheless, MPC's dependence on first-principles models limits its practical applicability, as uncertainties and disturbances commonly cause these models to become inaccurate or unreliable.

Neural networks can universally approximate any continuous and bounded function on a compact set [7], such as modeling disturbances on a quadrotor [8]. However, traditional learning-based approaches struggle to quickly adapt to sudden disturbances with limited data, which motivates us to explore the use of the adaptive neural network (ANN). To address this, we propose the ANNDO that adaptively compensates for disturbances within the MPC framework, enabling reliable and precise quadrotor control. The main contributions of this article can be summarized as follows:

1) An adaptive neural network disturbance observer is proposed to estimate disturbances, and the Lyapunov technique is employed to guarantee convergence of the estimation error.
2) Within the MPC framework, disturbance estimates are integrated into the predictive model, resulting in an ANNDO-enhanced MPC algorithm (ANN-MPC) that achieves robust trajectory tracking for quadrotors.
3) The effectiveness of our proposed approach is validated through extensive simulations and physical experiments.

B. Related Works

Trajectory tracking for quadrotor has emerged as a critical area of research, with extensive taxonomies provided in [4], [9]. Early approaches primarily relied on linearized controllers based on small-angle assumptions, which constrained their applicability to quasi-static hovering and low-acceleration maneuvers [10]. To overcome these limitations, nonlinear model predictive control has been extensively investigated due to its capability to generate optimal control inputs while simultaneously accounting for system constraints and future states. Empirical studies have demonstrated its superior performance in agile trajectory tracking [5], [6]. Nevertheless, its effectiveness is inherently dependent on

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accurate system modeling, rendering it susceptible to model uncertainties and external disturbances, such as wind perturbations [10]. These vulnerabilities present substantial challenges for real-world deployment, particularly in dynamically varying environments.

Various enhancement strategies have been developed to improve MPC's robustness against disturbances and model uncertainties. One widely adopted approach involves integrating disturbance observers to estimate and compensate for perturbations in real time. Techniques such as active disturbance rejection control (ADRC) [11], [12] have been extensively explored in this context. The extended state observer (ESO) is a key part of the ADRC, which treats internal unknown dynamics and external perturbations as the total perturbations of the extended state and estimates the total perturbations in real time [13]. Chen et al. [14] incorporated ESO within an MPC framework, leading to improved tracking accuracy by mitigating the impact of external disturbances.

Beyond disturbance observers, adaptive control has been increasingly integrated with other control frameworks to address uncertainties and enhance performance. In particular, L1 adaptive control has been integrated into MPC to enhance its robustness, primarily targeting the compensation of matched uncertainties [15], [16], [17]. However, these approaches are inherently limited when it comes to handling unmatched uncertainties, such as external forces projected onto the body-frame xy plane, which can significantly impair flight stability and tracking accuracy To address this issue, Chee et al. [18] introduced the KNODE-MPC-L1-Int framework, which synergistically combines KNODE for system modeling, L1 adaptive control for residual disturbance compensation, and MPC for trajectory optimization. This integrated approach has demonstrated notable improvements in robustness across varying environmental conditions.

Recent advancements in data-driven methodologies have further contributed to MPC's resilience against model uncertainties. Machine learning methods have made significant contributions to the approximation of dynamic residuals and the estimation of unknown disturbances [8], [19], [20], [21], [22], [23], [24], [25]. Gaussian process regression (GPR) has been successfully applied to capture unknown aerodynamic disturbances, resulting in enhanced MPC-based tracking accuracy [20]. Despite these advancements, the efficacy of machine learning-based models remains constrained by their reliance on extensive training data. Their generalization capability, particularly in unseen or dynamically shifting environments, remains an open challenge.

To mitigate the reliance of machine learning-based models on extensive precollected training datasets, ANN has been introduced as an alternative approach to real-time disturbance estimation [7], [26], [27], [28], [29]. By integrating the strong nonlinear approximation capability of neural networks with adaptive parameter adjustment, ANN can dynamically model complex disturbances and continuously refine their predictions based on real-time flight data [7]. However, despite these advantages, ensuring the boundedness of ANN weights remains a critical challenge [28], particularly when integrated with MPC. Stability concerns, alongside the computational overhead associated with real-time parameter updates, must be carefully addressed to ensure reliable deployment in practical quadrotor applications.

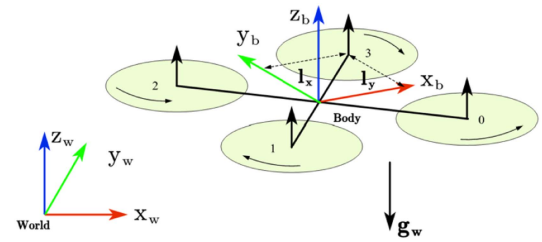

Fig. 1. Diagram of the quadrotor model with the world and body frames and propeller numbering convention.

II. Preliminary and Model

A. Notation

1) Kinetic Expression: As illustrated in Fig. 1, we define the world coordinate system W and the body-fixed frame B using the orthonormal basis $\{\boldsymbol{x}_W, \boldsymbol{y}_W, \boldsymbol{z}_W\}$ and $\{\boldsymbol{x}_b, \boldsymbol{y}_b, \boldsymbol{z}_b\}$. The body-fixed coordinate system is established at the center of mass of the quadrotor.

2) Neural Network Formulation: Consider a scalar input $\nu \in \mathbb{R}$ and a function $\varphi : \mathbb{R} \to \mathbb{R}$ which can be approximated by the following linear model:

$$\varphi(\nu) = \boldsymbol{W}^T \boldsymbol{h}(\nu) + \varepsilon, \tag{1}$$

where $\boldsymbol{h}(\nu) = [h_1(\nu), h_2(\nu), \ldots, h_n(\nu)]^T \in \mathbb{R}^n$ represents the vector of outputs from the hidden layer, $\boldsymbol{W} \in \mathbb{R}^n$ denotes the weight vector connecting the hidden units to the output, and $\varepsilon \in \mathbb{R}$ is the approximation error term. In this study, the Gaussian radial basis function neural network (RBFNN) is adopted as the activation function, defined by $h_i(\nu) = \exp(-\|\nu - \mu_i\|^2/\sigma^2) \in \mathbb{R}$, for $i = 1, 2, \ldots, n$, where the parameter $\sigma \in \mathbb{R}$ controls the spread of each basis function, and $\mu_i \in \mathbb{R}$ specifies the center position of the i-th Gaussian function.

Assumption 1: The approximation error ε is bounded in magnitude by an unknown constant $\psi \in \mathbb{R}$, such that $|\varepsilon| \leq \psi$.

B. Quadrotor Dynamics

The quadrotor dynamics can be modeled as:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\boldsymbol{p}} \\ \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{v}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v} \\ \frac{1}{2}\boldsymbol{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \\ (1/m)(\boldsymbol{R}_B^W(\boldsymbol{q})\boldsymbol{T} + \boldsymbol{d}_f) + \boldsymbol{g}_W \\ J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{d}_\tau) \end{bmatrix}, \tag{2}$$

where $\boldsymbol{p} \in \mathbb{R}^3$ denotes the position, and $\boldsymbol{v} \in \mathbb{R}^3$ represents the linear velocity in the world frame. The unit quaternion $\boldsymbol{q} = [q_w, q_x, q_y, q_z]^T$ represents the orientation of the body frame relative to the world frame. Furthermore, the angular velocity $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T \in \mathbb{R}^3$ is expressed in the body frame. Here, the scalar m denotes the mass, and $J = \mathrm{diag}(J_x, J_y, J_z)$ is the constant inertia matrix. The gravitational acceleration vector is defined as $\boldsymbol{g}_W = [0, 0, -9.81]^T\ \mathrm{m/s^2}$.The terms $\boldsymbol{d}_f \in \mathbb{R}^3$ and $\boldsymbol{d}_\tau \in \mathbb{R}^3$ denote the external disturbance force and torque acting

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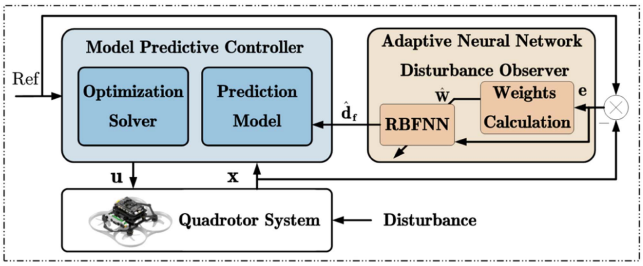

Fig. 2. The figure illustrates the integration of an adaptive neural network disturbance observer with model predictive control under disturbances. It describes the flow of data and control signals between these modules, highlighting the adaptive updating of the neural network weights, which enables precise and robust flight control despite disturbances.

on the quadrotor, resulting from aerodynamic drag, wind gusts, and other environmental influences.

The operator $\otimes$ denotes quaternion multiplication, which is explicitly defined as:

$$\boldsymbol{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \boldsymbol{q}. \tag{3}$$

The rotation matrix from the body frame to the world frame, $\boldsymbol{R}_B^W(\boldsymbol{q}) \in SO(3)$, is given by

$$\boldsymbol{R}_B^W(\boldsymbol{q}) = \begin{bmatrix} 1-2q_y^2-2q_z^2 & 2(q_xq_y-q_wq_z) & 2(q_wq_y+q_xq_z) \\ 2(q_xq_y+q_wq_z) & 1-2q_x^2-2q_z^2 & 2(q_yq_z-q_wq_x) \\ 2(q_xq_z-q_wq_y) & 2(q_wq_x+q_yq_z) & 1-2q_x^2-2q_y^2 \end{bmatrix}. \tag{4}$$

The collective thrust $\boldsymbol{T}$ and torque vector $\boldsymbol{\tau}$ in the body frame are computed as

$$\boldsymbol{T} = \begin{bmatrix} 0 \\ 0 \\ \sum_{i=0}^{3} T_i \end{bmatrix}, \quad \boldsymbol{\tau} = \begin{bmatrix} l_y(-T_0-T_1+T_2+T_3) \\ l_x(-T_0+T_1+T_2-T_3) \\ c_\tau(-T_0+T_1-T_2+T_3) \end{bmatrix}, \tag{5}$$

where l_x, l_y denote the effective distances from the center of mass to each rotor along the body axes, and c_τ is the torque coefficient relating the thrust to yaw torque. The individual rotor thrusts $\boldsymbol{u} = [T_0, T_1, T_2, T_3]^T \in \mathbb{R}^4$ are the control inputs.

Finally, the discrete-time state-space propagation using the explicit 4th-order Runge-Kutta (RK4) method is given by

$$\boldsymbol{x}_{k+1} = f_{\text{RK4}}(\boldsymbol{x}_k, \boldsymbol{u}_k). \tag{6}$$

III. Design of ANN-MPC Algorithm

This section presents an ANN-MPC algorithm. First, the ANNDO is designed to estimate the total disturbance. Then, the MPC is developed to track the desired trajectory while compensating for the total disturbance. The structure of the integrated controller and observer is shown in Fig. 2.

A. ANNDO Design

Consider the reduced state vector $\boldsymbol{z} = \boldsymbol{v}_W \in \mathbb{R}^3$, representing the world-frame velocity. Its dynamical evolution captures both nominal and uncertain system behaviors through the following differential equation:

$$\dot{\boldsymbol{z}} = f(.) + (1/m)\boldsymbol{d}_f, \tag{7}$$

where $f(.)$ is the nominal dynamics:

$$f(.) = (1/m)\boldsymbol{R}_B^W(\boldsymbol{q})\boldsymbol{T} + \boldsymbol{g}_W. \tag{8}$$

The observer of $\boldsymbol{z}$ is designed as:

$$\dot{\hat{\boldsymbol{z}}} = f(.) + (1/m)\hat{\boldsymbol{d}}_f + A_s\boldsymbol{e}, \tag{9}$$

where the estimation error is defined by $\boldsymbol{e} = \hat{\boldsymbol{z}} - \boldsymbol{z} \in \mathbb{R}^3$, with $A_s \in \mathbb{R}^{3\times 3}$ being the observer gain matrix, and $\hat{\boldsymbol{z}} \in \mathbb{R}^3$ the observer state.

Subtracting (7) from (9) yields the following error dynamics:

$$\dot{\boldsymbol{e}} = -A_s\boldsymbol{e} + (1/m)\left(\hat{\boldsymbol{d}}_f - \boldsymbol{d}_f\right). \tag{10}$$

For analytical convenience, the three-dimensional disturbance vector $\boldsymbol{d}_f = [d_x, d_y, d_z]^T \in \mathbb{R}^3$ is decomposed into its scalar components $d_x, d_y, d_z \in \mathbb{R}$. Each component is treated independently, allowing the error dynamics to be analyzed on a scalar basis. Specifically, let $d \in \{d_x, d_y, d_z\}$ denote an arbitrary scalar disturbance component, and correspondingly let $\hat{d}$ be its estimate. The scalar form of the error dynamics for each component $e \in \mathbb{R}$ is then expressed as

$$\dot{e} = -a_s e + (1/m)(\hat{d} - d), \tag{11}$$

where $a_s > 0$ is the scalar gain associated with the corresponding diagonal entry of the gain matrix A_s.

According to (1), the true disturbance $d = \boldsymbol{W}^T\boldsymbol{h}(e) + \varepsilon$ is approximated by the neural network, and its estimate is given by

$$\hat{d} = \hat{\boldsymbol{W}}^T\boldsymbol{h}(e), \tag{12}$$

where $\hat{\boldsymbol{W}}$ denotes the estimate of the optimal weight vector $\boldsymbol{W}$.The scalar error dynamics can now be expressed in terms of weight estimation error $\tilde{\boldsymbol{W}} = \hat{\boldsymbol{W}} - \boldsymbol{W}$:

$$\dot{e} = -a_s e + (1/m)\left(\tilde{\boldsymbol{W}}^T\boldsymbol{h}(e) - \varepsilon\right). \tag{13}$$

The adaptive update law is designed as

$$\dot{\hat{\boldsymbol{W}}} = -(1/m)\gamma e\,\Phi\,\boldsymbol{h}(e), \tag{14}$$

where $\gamma \in \mathbb{R}$ is a positive adaptation gain and $\Phi \in \mathbb{R}$ is a positive weight gain.

Theorem 1: Consider the closed-loop system described by equations (7)–(13) with the adaptive law (14). Assume that Assumption 1 strictly holds and the matrix A_s is positive definite. The observer estimation error e and the parameter estimation error $\tilde{\boldsymbol{W}}$ are uniformly ultimately bounded (UUB).

Proof: Consider the Lyapunov function candidate

$$V = \frac{1}{2}e^T\Phi e + \frac{1}{2\gamma}\tilde{\boldsymbol{W}}^T\tilde{\boldsymbol{W}}, \tag{15}$$

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where Φ satisfies the equation $a_s^T\Phi + \Phi a_s = -Q$, with $Q = Q^T > 0 \in \mathbb{R}$.

Taking derivative of V along the trajectories yields, and substituting $\dot{e}$ and $\dot{\hat{\boldsymbol{W}}}$, yielding

$$\dot{V} = -\frac{1}{2}e^TQe + (1/m)\Big(e^T\Phi\tilde{\boldsymbol{W}}^T\boldsymbol{h}(e) + e^T\Phi\varepsilon - \tilde{\boldsymbol{W}}^Te\Phi\boldsymbol{h}(e)\Big). \tag{16}$$

Note that $e^T\Phi\tilde{\boldsymbol{W}}^T\boldsymbol{h}(e)$ and $\tilde{\boldsymbol{W}}^Te\Phi\boldsymbol{h}(e)$ are equal scalars and thus exactly cancel each other in the Lyapunov derivative, yielding

$$\dot{V} = -\frac{1}{2}e^TQe + (1/m)e^T\Phi\varepsilon. \tag{17}$$

Since $Q > 0$, the first term is negative definite. According to Assumption 1, the second term is bounded by $|(1/m)e^T\Phi\varepsilon| \leq (1/m)\|e\|\|\Phi\|\|\psi\|$.

Since the Lyapunov function accounts for both the estimation error e and the parameter error $\tilde{\boldsymbol{W}}$, and $\dot{V}$ is negative semi-definite outside a compact set due to bounded approximation error ε, standard Lyapunov theory ensures e is UUB.

Furthermore, the adaptation law $\dot{\hat{\boldsymbol{W}}} = -(1/m)\gamma e\,\Phi\,\boldsymbol{h}(e)$, driven by bounded e and bounded basis functions $\boldsymbol{h}(e)$, ensures that $\tilde{\boldsymbol{W}}$ is also UUB.

Thus, both the observer error and weight estimation error remain bounded, guaranteeing closed-loop stability and parameter convergence within a bounded neighborhood. This completes the proof.

Note: The proposed observer framework is theoretically capable of estimating both disturbance forces and torques. In this work, we focus on estimating and compensating for the disturbance forces. However, the disturbance torque estimation capability can be systematically realized by reformulating the nominal translational dynamics in equation (7) into corresponding rotational dynamics equations.

B. ANN-MPC Design

To achieve robust quadrotor trajectory tracking under disturbances, the MPC incorporating disturbance estimates from the ANNDO is proposed. We summarize the proposed method in Algorithm 1.

During each iteration of solving the MPC problem, the estimated disturbance force $\hat{\boldsymbol{d}}_k$ is assumed to remain constant along the prediction horizon. The nonlinear trajectory tracking problem is formulated as a quadratic optimization minimizing the weighted sum of state and control tracking errors over a discrete horizon T with N steps:

$$\begin{aligned} \min_{\boldsymbol{u}} \quad & \Delta\boldsymbol{x}_N^T\boldsymbol{Q}\Delta\boldsymbol{x}_N + \sum_{k=0}^{N-1}\left(\Delta\boldsymbol{x}_k^T\boldsymbol{Q}\Delta\boldsymbol{x}_k + \Delta\boldsymbol{u}_k^T\boldsymbol{R}\Delta\boldsymbol{u}_k\right) \\ \text{s.t.} \quad & \boldsymbol{x}_{k+1} = f_{\text{RK4}}(\boldsymbol{x}_k, \boldsymbol{u}_k), \\ & \boldsymbol{x}_0 = \boldsymbol{x}_{\text{init}}, \boldsymbol{u}_k \in \mathbb{U}, \boldsymbol{x}_k \in \mathbb{X}, \boldsymbol{x}_N \in \mathbb{X}_N, \end{aligned} \tag{18}$$

where $\Delta\boldsymbol{x}_k = \boldsymbol{x}_k - \boldsymbol{x}_{k,r}$ and $\Delta\boldsymbol{u}_k = \boldsymbol{u}_k - \boldsymbol{u}_{k,r}$ denote the tracking errors relative to the reference trajectory. The weighting matrices $\boldsymbol{Q} \geq 0$ and $\boldsymbol{R} > 0$ penalize tracking deviations and control effort, respectively. The sets $\mathbb{U}$, $\mathbb{X}$, and $\mathbb{X}_N$ represent admissible input, state, and terminal constraint sets, respectively. This optimization problem is solved online using a real-time sequential quadratic programming (SQP) method [30] implemented with the ACADO toolkit [31]. The proof of the recursive feasibility of the proposed MPC and the stability of the combined system is provided based on the work in [32].

Algorithm 1: ANN-MPC Algorithm.

Input: State measurements $\boldsymbol{x}_k$, reference trajectory $\boldsymbol{x}_{k,r}$
Output: Control command $\boldsymbol{u}_k^*(0)$
Initialize: $\boldsymbol{u}_k^* \leftarrow \boldsymbol{0}$, $\hat{\boldsymbol{d}}_k \leftarrow \boldsymbol{0}$, $\hat{\boldsymbol{W}} \leftarrow \boldsymbol{0}$, $\boldsymbol{x}_0$ given
for each time step k **do**
  Update the ANNDO weights $\hat{\boldsymbol{W}}$ using (14)
  Compute disturbance estimate $\hat{\boldsymbol{d}}_k$ using (12)
  Build prediction model by (6)
  Apply $\boldsymbol{x}_k$ to solve (18) and obtain the optimal control sequence
  $\boldsymbol{u}_k^* = \{\boldsymbol{u}_k^*(0), \boldsymbol{u}_k^*(1), \ldots, \boldsymbol{u}_k^*(N-1)\}$
  Apply the first control input $\boldsymbol{u}_k^*(0)$ to the system
end for

IV. Experiments and Results

During testing, we address the following questions:

1) What is the disturbance estimation performance of ANNDO?
2) How much does ANN-MPC tracking performance improve in the presence of gusts, unknown loads, and aerodynamic drag?
3) Can simulation-based tracking improvements be validated through flight tests?

We address these questions through a series of simulations and physical tests, incorporating significant disturbances, and further illustrate the algorithm's robustness with the corresponding video.

To validate the effectiveness of the proposed method, we conduct comparative studies against four benchmark approaches: MPC, Gaussian Process Regression-based MPC (GP-MPC) [20], Extended State Observer-based MPC (ESO-MPC) [14], and L1 Adaptive Control-integrated MPC (L1-MPC-Int) [18]. All approaches use the same MPC parameters as ANN-MPC, ensuring a fair comparison. Performance is assessed by comparing the root-mean-square error (RMSE) between the quadrotor's position and the desired trajectory.

A. Simulation Setup

We employ the Python-based simulation framework from [20] for preliminary validation of the proposed algorithm. The system dynamics are numerically integrated using a 4th-order explicit Runge-Kutta method with a fixed time step of 0.5 ms. The simulation environment assumes precise quadrotor odometry, accurate tracking of the commanded single-rotor thrusts, and instantaneous computation of the MPC. The simulator also

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TABLE I
COMPARISON OF ACCELERATED CIRCULAR TRAJECTORY TRACKING PERFORMANCE AT DIFFERENT PEAK VELOCITIES IN SIMULATION

	MPC	GP-MPC		ESO-MPC		L1-MPC-Int		ANN-MPC	
v_{peak} [m s^{-1}]	RMSE [m]	RMSE [m]	%↓	RMSE [m]	%↓	RMSE [m]	%↓	RMSE [m]	%↓
2	0.0478	0.0139	70.92	0.0084	82.43	0.0048	89.96	0.0038	**92.05**
4	0.1164	0.0410	64.78	0.0368	68.38	0.0253	78.26	0.0180	**84.54**
6	0.1821	0.0654	64.09	0.0652	64.20	0.0453	75.12	0.0318	**82.54**
8	0.2407	0.0875	63.65	0.0916	61.94	0.0642	73.33	0.0437	**81.84**
10	0.2925	0.1092	62.67	0.1149	60.72	0.0809	72.34	0.0533	**81.78**
12	0.3395	0.1313	61.33	0.1388	59.12	0.0978	71.19	0.0624	**81.62**

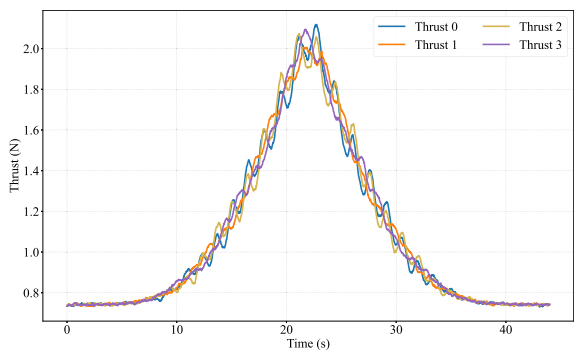

Fig. 3. Thrust commands of proposed method during accelerated circular trajectory with a peak velocity of 10 m/s in simulation test.

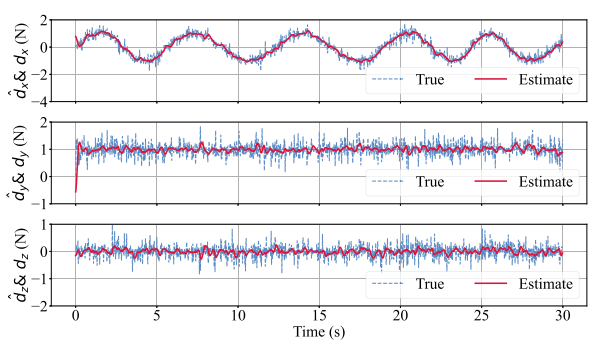

Fig. 4. Comparison of disturbance estimated by ANNDO and real disturbance.

accounts for rotor and airframe aerodynamic drag effects. We designed a set of experiments as follows:

1) Agile Flight Capability Test in Simulation: Tracking performance of ANN-MPC was compared and analyzed against several other controllers under aggressive flight trajectories. The effect of increasing speed on tracking accuracy was simulated on circular trajectories with a radius of 5 m, where peak speeds varied between 2 and 12 m/s. In these cases, the mass, inertia, and rotor arm lengths of the quadrotors were perfectly known. These experiments provide insights into the maximum performance achievable by each method in the presence of an unmodeled resistance model.

As shown in Table I, each row represents the cumulative positional tracking error at a given peak speed. As flight speed

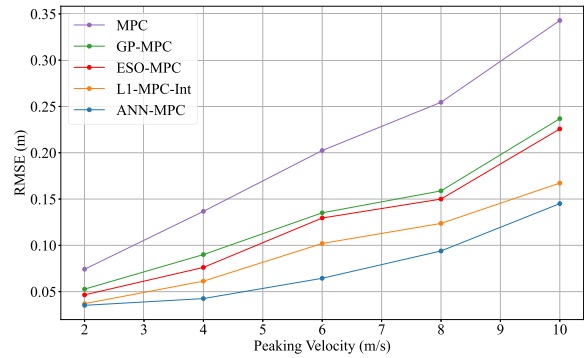

Fig. 5. Tracking performance comparison.

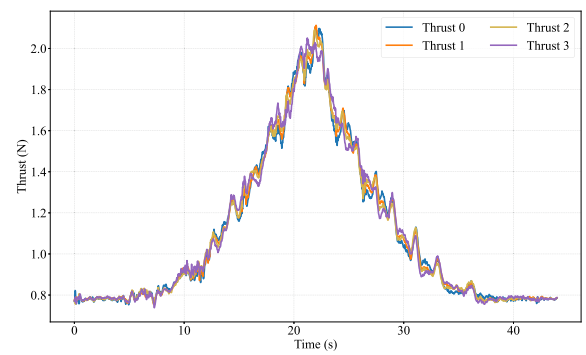

Fig. 6. Thrust commands of proposed method during accelerated circular trajectory with a peak velocity of 10 m/s in physical test.

increases, the proposed ANN-MPC maintains accurate trajectory tracking, with significantly higher accuracy compared to other methods. This demonstrates that ANN-MPC effectively compensates for unmodeled resistance, enabling agile flight. The simulation results in Fig. 3 demonstrate that the proposed ANN-MPC algorithm generates smooth thrust commands within all motors' saturation limits while exhibiting rapid, stable response characteristics, thereby verifying its theoretical feasibility.

2) Disturbance Estimate Capability Test: To validate the external disturbance estimation capability of ANNDO, $\boldsymbol{f}_{\text{total}}(t) = [\sin(\omega t), 1, 0]^T + \boldsymbol{\epsilon}(t)$ is designed to model unknown disturbance, where ω is the angular frequency of the periodic disturbance component, and $\boldsymbol{\epsilon}(t) = N(0, \sigma^2(t)) \cdot [1, 1, 1]^T \in \mathbb{R}^3$ denotes a noise disturbance modeled by a zero-mean Gaussian distribution.

The comparison between the actual disturbance and the disturbance estimated by the ANNDO is shown in Fig. 4. The results indicate that the estimated disturbance converges rapidly and exhibits excellent tracking performance, validating the effectiveness of the proposed ANNDO in accurately estimating disturbances.

B. Physical Experiments Setup

Physical experiments were carried out using a custom-built quadrotor with a total mass of 250 g. The vehicle was equipped with a propulsion system comprising four Smoox 1404 brushless motors paired with 6045 three-bladed propellers, yielding

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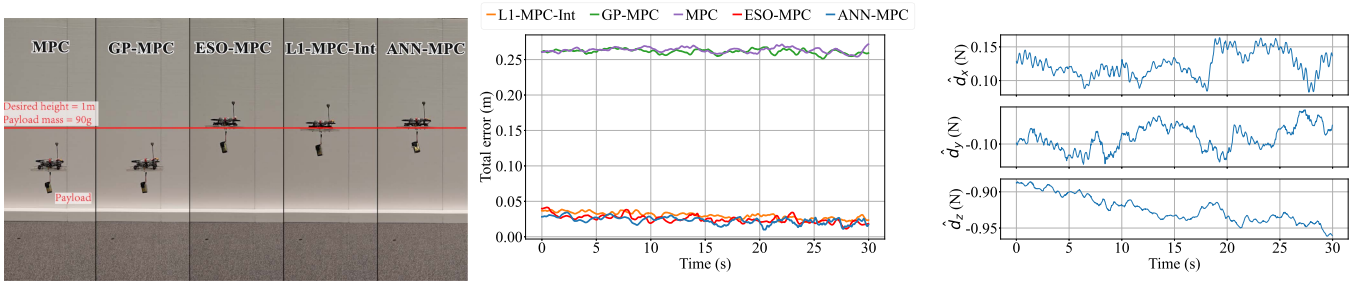

Fig. 7. The results of the hovering with unknown load experiment. Left: Experimental setup for hovering with an unknown load, using a 90 g battery as the suspended load. Center: Tracking error during hovering with the load. Right: Real-time disturbance estimation using ANNDO.

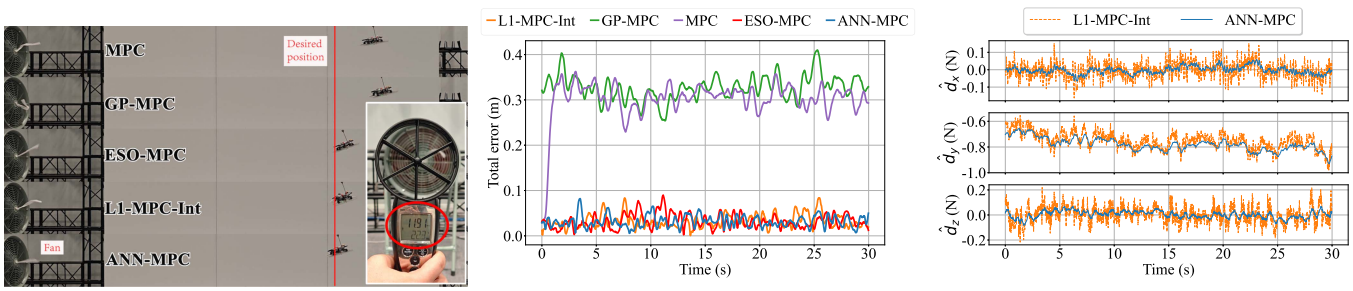

Fig. 8. The results of the hovering anti-wind disturbance experiment: Left: Experimental setup for hovering in wind. Center: Tracking error during wind hover. Right: Comparison of disturbance estimation between ANN-MPC and L1-MPC-Int.

a maximum thrust-to-weight ratio of 4:1 to ensure adequate maneuverability. The quadrotor's embedded system is based on a custom-designed flight controller board running PX4 firmware, employing the STM32H743 microcontroller.

The localization is achieved using an OptiTrack motion capture system with 46 infrared cameras, providing high-precision full-state estimation, including position, velocity, attitude, and angular velocity, within a $20 \times 10 \times 10$ m^3 experimental space. The proposed algorithm operates at 50 Hz on a laptop equipped with an Intel i7 processor, implementing a prediction horizon of $N = 20$. Control commands, including collective thrust and desired body rates, are transmitted to the quadrotor via an LR24 wireless module, while the onboard PID controller handles low-level command tracking. The software architecture is based on the Robot Operating System (ROS). To evaluate the practical effectiveness of the proposed algorithm, we conducted the following tests:

1) Agile Flight Capability Test in Physical Flight: Performance comparison of various algorithms was conducted during the tracking of an accelerated circular trajectory. The experimental configuration mirrored that of Simulation Experiment 1, featuring a circular path with a radius of 4 meters, while the peak velocity progressively increased from 2 m/s to 10 m/s.

As shown in Fig. 5, the tracking accuracy of ANN-MPC is significantly improved by over 60% compared to the nominal MPC, demonstrating that the ANN-MPC method effectively compensates for unmodeled aerodynamic drag residuals, outperforming other disturbance compensation strategies. The thrust commands in the physical test are shown in Fig. 6. The results are highly consistent with the simulation results, which further verifies the feasibility and effectiveness of the proposed algorithm deployed in the real world.

2) Robustness to Mass Mismatch in Physical Flight: To validate the robustness of the proposed algorithm in the presence of unknown loads, we added a 90 g payload to the quadrotor to simulate model uncertainties. As shown in Fig. 7, the GP-MPC and MPC methods exhibit significant oscillations and large steady-state errors under mass mismatch. In contrast, the ANN-MPC method enhances steady-state accuracy to within 3 cm, reducing the error by approximately 24 cm compared to the MPC, representing an 88.89% relative improvement. This demonstrates that the ANN-MPC method can quickly adapt to mass mismatch and significantly enhance system stability.

In addition, this experiment primarily validates the estimation and compensation of matched disturbances along the Z-axis of the quadrotor. Notably, during flight, as the battery voltage decays, the ANN method promptly detects and effectively compensates for disturbances caused by changes in voltage. These results highlight the exceptional robustness and adaptability of the ANN-MPC in dynamic environments.

3) Robustness to Wind Disturbance During Hovering: We further validated the robustness of the proposed algorithm through hover flight in windy conditions. The wind, generated by an industrial fan, directly impacted the Y-axis of the quadrotor during hover, with the maximum wind speed at the hover position reaching approximately 12 m/s, equivalent to a force 5 wind on the Beaufort scale.

Fig. 8 shows a comparison of the real-time total error for various methods. Compared to the uncompensated MPC and

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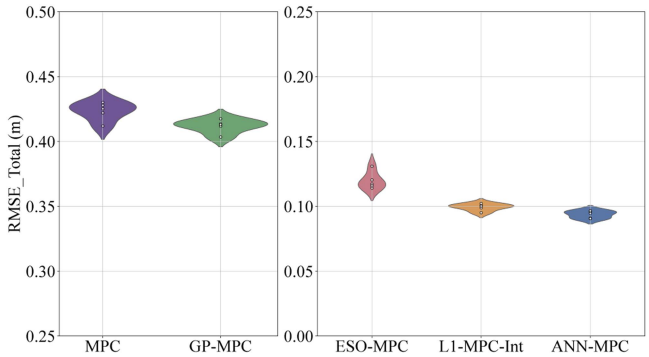

Fig. 9. Violin plot of position tracking RMSE distributions from complex disturbance flight experiments, comparing the performance of different methods.

the GP-MPC method, the steady-state error of the ANN-MPC method was reduced by approximately 25 cm. This indicates that the ANN-MPC method can effectively estimate and compensate for time-varying external disturbances. Compared to the L1-MPC-Int method, the ANN-MPC not only accurately estimates the disturbance but also produces very low disturbance noise, demonstrating excellent estimation accuracy. Furthermore, the ANN-MPC method requires no additional filtering and can be directly integrated with the dynamic equations of the quadrotor, enhancing its robustness.

4) Robustness to Complex Disturbances: In order to assess the robustness of the proposed method under more complex conditions, various disturbances were introduced, including gusts, unknown loads, and unmodeled aerodynamic effects. The task of the quadcopter is to fly along a circular trajectory with a radius of 3 meters at a speed of 2 m/s. Throughout the experiment, three industrial fans generated a wind field within a radius of 3 meters, producing a maximum wind speed of 12 m/s. In addition, a 90 g mass is attached to simulate the effects of unknown load interference.

Each method was tested in five flight experiments, and the RMSE distribution of position tracking is summarized in Fig. 9. The results indicate that, in the presence of complex disturbances, the proposed method exhibits the lowest concentration of error distribution, with tracking performance at least 50% better than MPC and at least 15% better than L1-MPC-Int. This demonstrates that the ANN-MPC method has exceptional adaptability to complex and unmodeled external disturbances, such as gusts and load variations.

5) Outdoor Experiments: As demonstrated in Fig. 10, the quadrotors successfully tracked an accelerated 3D lemniscate trajectory under outdoor windy conditions with wind speeds of approximately 5 m/s, and the corresponding tracking performance is presented in Fig. 11. Under positioning discrepancies conditions, the ANN-MPC achieved an RMSE less than 15 cm, representing an improvement exceeding 60% compared to the baseline MPC. Notably, while the RMSE demonstrated an increasing trend with higher flight speeds and more aggressive trajectories along the green segment, the overall tracking error of ANN-MPC remained at a low level. These results confirm the adaptability and robustness of ANN-MPC in challenging outdoor wind environments.

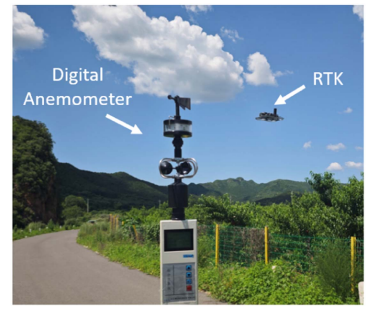

Fig. 10. Experimental scene in outdoor environments under strong wind conditions.

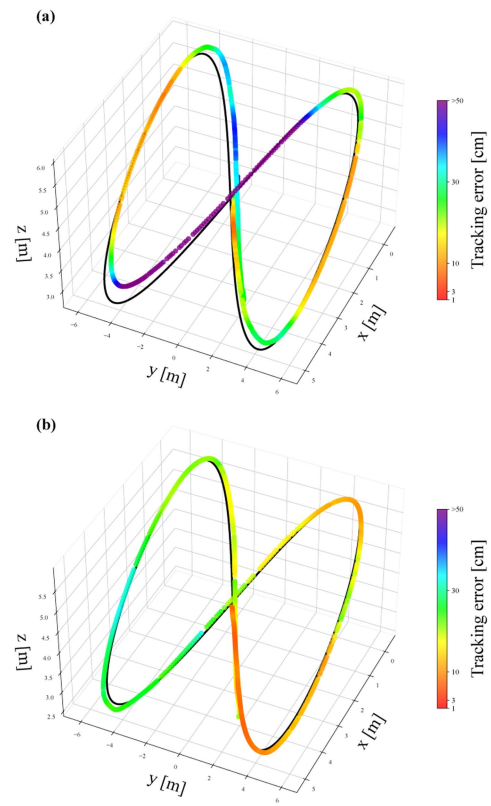

Fig. 11. Comparison of quadrotor tracking performance under different control methods:(a)MPC, RMSE = 0.4112 m; (b)ANN-MPC, RMSE = 0.1495 m.

V. Conclusion

This letter proposes an ANN-MPC algorithm for robust trajectory tracking of quadrotors under disturbance conditions. First, ANNDO is designed for real-time disturbance estimation, ensuring convergence of the estimation error using Lyapunov-based methods. Subsequently, the disturbance estimated by ANNDO is integrated into the predictive model to construct an observer-based MPC. The efficacy and practicality of the approach have been thoroughly validated through both simulation studies and real-world flight experiments. Future work will extend the adaptation framework to rotational dynamics and investigate disturbance prediction models using historical estimation data, potentially enhancing MPC prediction accuracy for aggressive maneuvers.

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